

Malek Hassan

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EDUCATION

CODE University of Applied Sciences <i>B.Sc. Software Engineering - final year</i>	2026 - Present <i>Berlin, Germany</i>
Transferred from RWTH Aachen to finish the degree in Berlin's project-based, ship-real-software curriculum.	
RWTH Aachen University <i>B.Sc. Computer Science - first two years</i>	Oct 2023 - 2026 <i>Aachen, Germany</i>

EXPERIENCE

African Arab International Bank <i>Software Engineering Intern, Digital Factory</i>	Jul 2025 - Sep 2025 <i>Cairo, Egypt</i>
- Built an end-to-end banking support assistant with HTML, CSS, JavaScript/TypeScript, URL-scraped public knowledge, and locally hosted Ollama inference. - Won adoption of the concept by the development team and documented its retrieval, inference, and data flow for the bank's internal review process.	
RWTH Aachen University <i>Student Technical Support, CS Peer Assistance</i>	Oct 2023 - Jan 2024 <i>Aachen, Germany</i>
Supported peers with Java debugging, Git/GitHub workflows, and LaTeX document structure.	

SELECTED PROJECTS

MOZCODE - Symbol-Level Context Engine for Coding Agents <i>Creator and maintainer MIT licensed</i>	2026 <i>github.com/Malek1414/MOZCODE</i>
- Built a local MCP server for Claude Code, Codex, and OpenAI models that uses tree-sitter ASTs to return requested symbols and outlines instead of entire files, with grouped search, AST-anchored edits, and read-only SQLite/PostgreSQL schema introspection. - Instrumented an append-only JSONL ledger and self-contained SVG dashboard; measured an estimated 2,180,966 input tokens avoided across 1,159 optimized calls and 16 projects, with 80% average payload reduction versus disclosed whole-file/plain-grep baselines.	
FollowCam - Robot Cameraman and Game Analytics <i>Winner, CODE Hackathon Berlin</i>	Aug 2026 <i>Swift YOLOv8 Arduino</i>
- First place with a two-person team: a 3D-printed ring-and-fork mount on a metal-gear servo turns any tripod into a ~€20 auto-tracking camera, steered by on-device vision from an iOS app. - Fine-tuned YOLOv8 pipeline tracked players on real league footage, read jersey numbers, mapped the play to a 2D court, and computed shots, points, and possession - attempt detection human-checked at 96% precision.	
Second Brain - End-to-End RAG Knowledge System <i>Personal knowledge capture and cited retrieval</i>	2025 <i>Node.js ChromaDB Claude API</i>
Built one retrieval loop from voice/text capture through embedding and ChromaDB indexing to an Obsidian vault, queried through the Animus chat interface with citations back to the source notes; a native SwiftUI iOS app now captures and transcribes voice notes into the same loop.	

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, Java, SQL, C, HTML/CSS
AI and systems: RAG, Model Context Protocol, tree-sitter, vector search, prompt engineering, Claude API, Ollama
Backend and data: Node.js, Express, PostgreSQL, SQLite, Supabase, ChromaDB, REST APIs
Frontend and visualization: React, Next.js, Vite, D3.js, Recharts, Tailwind CSS, Canvas, Framer Motion
Developer tooling: Git/GitHub, Vitest, esbuild, PDF processing, command-line workflows

CERTIFICATIONS

Anthropic: Model Context Protocol (Introduction + Advanced Topics); Agent Skills; Subagents (May 2026)

LEADERSHIP

Organising Committee Head, Model United Nations <i>High school leadership role</i>	Sep 2022 - Jun 2023 <i>Cairo, Egypt</i>
- Led the organising committee across sales, marketing, logistics, and administration, and delivered a keynote to an audience of 300+. - Improved the event's operational performance by 20% year over year.	

SOCIAL

SummerUP Berlin - Team Offerprofi <i>One week of building and selling - top 10 of 72 teams</i>	Aug 2026 <i>offerprofi.de github.com/Malek1414/Offerprofi</i>
Built the AI layer, whose extraction schema cannot invent a price - every line item is quoted from the source document or flagged uncertain.	

AWARDS AND HONORS

Winner, CODE Hackathon Berlin (Aug 2026) - first place with FollowCam, a ~€20 auto-tracking robot cameraman with a live game-analytics pipeline, built and pitched as a two-person team.
State Champion, Basketball - won a national/state championship title in the youth division.

MOZCODE figures are estimates against local counterfactual baselines, not billing data. Snapshot: 29 Aug 2026.